

2025 ANNUAL GREENHOUSE GAS & CRITERIA AIR POLLUTANTS REPORT

Comprehensive Operational & Statutory Compliance Quantification

TOTAL SCOPE 1 & 2 EMISSIONS 1,908,700 tCO2e -15.9% vs Baseline	TOTAL FLARED VOLUME 117.9 MMSm3 Lowest Flaring on Record	FLARING INTENSITY (HBNS) 0.865% of Gas COMPLIANT (<1.00% Decree 21-330)
MEASURED FLARING DRE 99.89% (HBNS) / 99.85% (El Merk) VISR Camera Optical Verification	SALEABLE CARBON INTENSITY 33.72 kg CO2e / Saleable BOE -6.2% YoY Decarbonization	NGSI METHANE INTENSITY 0.043 wt.% Industry Top Decile (<0.20% Target)

Reporting Entity:	Groupement Berkine (Association Sonatrach / Occidental / Eni / TotalEnergies / Pertamina / Repsol)
Operating Assets:	Hassi Berkine South (HBNS, Block 404a) & El Merk (Block 208)
Reporting Period:	January 1, 2025 – December 31, 2025 (Annual Fiscal Cycle)
Verification Standard:	ISO 14064-1:2018 / GHG Protocol Corporate Standard / OGMP 2.0 (Level 4/5)
Statutory Regimes:	Algerian Executive Decree 21-330 (Gas Flaring) & Executive Decree 06-138 (Air Quality)
Calculation Basis:	API Compendium of GHG Emissions Methodologies for Oil & Gas (2021) / Site-Specific Gas Chromatography
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CHAPTER —

LEGAL STATEMENTS & MRV GOVERNANCE

Institutional Disclaimers and GHG Accounting Principles

MRV Governance & Compliance Framework

This document constitutes the quantified annual greenhouse gas (GHG) and criteria air pollutant (CAP) inventory for Groupement Berkine assets HBNS and El Merk. Compiled under ISO 14064-1:2018 specifications, API Compendium (2021), and national decrees 21-330 and 06-138.

1. Cautionary & Forward-Looking Statements

This report contains forward-looking statements concerning operational decarbonization trajectories, flare reduction projects, and methane abatement initiatives through 2030. Actual outcomes may differ depending on facility modifications, reservoir characteristics, gas re-injection availability, and technological maturation. Statements are made based on 2025 verified engineering metering and custody transfer data.

2. Quantification Principles & Hierarchy (ISO 14064-1:2018)

Principle	Application in Groupement Berkine 2025 Inventory
Relevance	All direct operational emission sources, flared streams, process vents, and fugitive components across HBNS and El Merk are accounted for without arbitrary omissions.
Completeness	Quantification includes Scope 1 stationary combustion, routine/non-routine/safety flaring, cold venting, and equipment leaks. Exclusions are formally justified under ISO 14064-1 de minimis rules.
Consistency	Time-series comparability is maintained from 2021 to 2025 using fixed high-heating value stoichiometric equations and continuous custody transfer metering.
Accuracy	Site-specific fuel gas chromatography (molecular weight, C1-C6+ speciation, H2S) is applied directly to combustion and flaring equations, achieving Level 4/5 OGMP 2.0 accuracy.
Transparency	All activity data, emission factors, continuous VISR camera flare combustion efficiency evaluations, and stack testing logs are maintained on tamper-evident digital ledgers.

3. 2030 Decarbonization Milestones & Target Alignment

Key Performance Metric	Baseline (2020)	2025 Actual	2030 Target	Target Status / Trajectory
Gross Scope 1 GHG Emissions	2,269,600 tCO2e	1,908,700 tCO2e	1,700,000 tCO2e	ON TRACK (-15.9% achieved vs 2020)
Operational Flaring Volume	193.0 MMSm3	117.9 MMSm3	Zero Routine Flaring	ON TRACK (-38.9% reduction achieved)
Flaring Intensity (% Gross Gas)	2.19% of gas	0.91% of gas	< 0.50% of gas	COMPLIANT (<1.00% Decree 21-330 ceiling)
Methane Flaring Efficiency (DRE)	98.0% (Default)	99.89% (VISR)	"e 99.50 % Verified"	ACHIEVED via Continuous Optical Tracking
Methane Intensity (NGSI wt.%)	0.165 wt.%	0.043 wt.%	< 0.050 wt.%	EXCEEDED (Top Decile OGCI Benchmarking)

CHAPTER 1 & 2

ORGANIZATIONAL BOUNDARIES & PRODUCTION PROFILE

Asset Geography, Operator Equity Structure, and Hydrocarbon Production Streams

Operational Parameter	HBNS Facility	El Merk Facility
Concession Block	Block 404a (Berkine Basin)	Block 208 (Berkine Basin)
Geographic Location	31° 42' N, 7° 12' E (Ouargla)	30° 48' N, 8° 42' E (Illizi)
Boundary Approach	Operational Control & Equity Share (51% Sonatrach)	Operational Control & Equity Share (51% Sonatrach)
Operator	Sonatrach / Groupement Berkine	Sonatrach / Groupement Berkine
Main Processing Units	Oil-gas separation, gas compression, water injection	Gas reinjection, LPG recovery, crude stabilization
Liquid Capacity	150,000 bopd crude oil stabilization	98,000 bopd liquids + 31,000 bpd LPG
Gas Handling Capacity	18.5 MMSm3/d gas compression & reinjection	16.0 MMSm3/d high-pressure gas reinjection

Chapter 4: Hydrocarbon Production Ledger (Fiscal Year 2025)

Product Stream	Unit	HBNS (Block 404a)	El Merk (Block 208)	Consolidated Total
Gross Gas Produced	MMSm3	5,517.62	4,239.89	9,757.51
Gas Reinjecte	MMSm3	5,124.93	3,865.08	8,990.01
Gas Produced w/o Injection	MMSm3	392.69	374.81	767.50
Crude Oil Production	kbbl	23,360.00	21,450.00	44,810.00
Condensates Production	kbbl	0.00	6,820.00	6,820.00
LPG Production	ktonnes	0.00	625.00	625.00
Total Liquid Production	MMBOE	23.36	32.02	55.38
Total Production (Gross BOE)	MMBOE	70.16	60.77	130.93
Total Production w/o Injected Gas	MMBOE	27.91	33.36	61.27
Total Saleable Production	MMBOE	25.44	31.18	56.62

Field Re-injection Intensity (>92%)

A remarkable 92.1% of all gross gas produced at HBNS and 91.2% at El Merk is reinjected for secondary reservoir pressure maintenance, drastically mitigating prospective venting and flaring while maximizing ultimate liquids recovery.

CHAPTER 5

QUANTIFIED GREENHOUSE GAS INVENTORY (2025)

Scope 1 & Scope 2 Modular Quantification per API Compendium & ISO 14064-1:2018

Modular Process Source	HBNS (tCO2e)	El Merk (tCO2e)	Consolidated (tCO2e)	% Share of Scope 1
Stationary Combustion (Turbines/Heaters)	662,100.00	947,300.00	1,609,400.00	84.32%
Operational Flaring (Routine/Non-Routine/Safety)	118,500.00	171,900.00	290,400.00	15.21%
Process Venting & Glycol Dehydrators	2,700.00	3,200.00	5,900.00	0.31%
Fugitive Equipment Leaks (Valves/Flanges)	1,500.00	1,500.00	3,000.00	0.16%
Total Scope 1 Direct Emissions	784,800.00	1,123,900.00	1,908,700.00	100.00%
Scope 2 Indirect Emissions (Purchased Grid)	0.00	0.00	0.00	0.00% (Self-Powered)
Total Scope 1 & Scope 2 Emissions	784,800.00	1,123,900.00	1,908,700.00	100.00%

Greenhouse Gas Breakdown by Molecule (IPCC AR5 100-yr GWP)

Gas Species	Molecular Formula	Mass Emitted (Tonnes)	GWP Factor (AR5)	Emissions (tCO2e)	% of Inventory
Carbon Dioxide	CO2	1,811,940.00	1.0	1,811,940.00	94.93%
Methane	CH4	3,360.00	28.0	94,080.00	4.93%
Nitrous Oxide	N2O	10.11	265.0	2,680.00	0.14%
Total Quantified GHG Footprint	CO2e	—	—	1,908,700.00	100.00%

Self-Generated Electrical Energy (Zero Scope 2)

Both HBNS and El Merk operate as autonomous island power systems using on-site gas turbines fed by processed fuel gas. No grid power is imported from Sonelgaz, rendering Scope 2 emissions exactly 0.00 tCO2e.

CHAPTER 6

OPERATIONAL FLARING & EXECUTIVE DECREE 21-330

Stream Disaggregation, Optical DRE Verification, and Statutory Limit Scorecard

Executive Decree 21-330 Article 9 Statutory Mandate

Article 9 of Algerian Executive Decree 21-330 establishes that total flared gas must not exceed 1.00% of gross gas production. HBNS has achieved full compliance at 0.865%, while El Merk recorded 1.65% with committed flare gas recovery CAPEX scheduled for 2026 completion.

Operational Flaring Metric	Unit	HBNS	El Merk	Consolidated Total
Flaring Parameter	Unit	HBNS (Block 404a)	El Merk (Block 208)	Consolidated Total
Routine Flaring Volume	kNm3	26,740.00	47,246.00	73,986.00
Non-Routine Flaring (Process Upsets)	kNm3	19,100.00	17,269.00	36,369.00
Safety, Purge & Pilot Gas	kNm3	1,910.00	5,633.00	7,543.00
Total Flared Volume	kNm3	47,750.00	70,148.00	117,898.00
Total Flared Volume in MMSm3	MMSm3	47.75	70.15	117.90
Gross Gas Produced	MMSm3	5,517.62	4,239.89	9,757.51
Flaring Intensity (% Gross Gas)	vol.%	0.865%	1.654%	1.208%
Statutory Flaring Ceiling (Decree 21-330)	vol.%	1.000%	1.000%	1.000%
Statutory Compliance Status	Decree 21-330	COMPLIANT	EXCEEDED	PARTIAL COMPLIANCE
Measured Flare DRE (VISR Camera)	%	99.89%	99.85%	99.87% (Avg)
DRE Measurement Method	Method	Optical Multispectral Camera	Optical Multispectral Camera	Continuous Empirical VISR

Key Flare Reduction Milestones Achieved in 2025:

- HBNS Lowest Flaring on Record: 47.75 MMSm3 flared in 2025 represents a 42% drop compared to 2021 levels.
- VISR Optical Camera DRE: Measured Destruction and Removal Efficiency of 99.89% confirms that over 99.8% of methane is destroyed, far surpassing the conservative 90% target.
- El Merk Mitigation Roadmap: Low-pressure flare gas recovery compressors (FGRS) are scheduled for commissioning in Q3 2026 to bring El Merk flaring below the 1.00% statutory limit.

CHAPTER 7

CRITERIA AIR POLLUTANTS & EXECUTIVE DECREE 06-138

Stack Concentration Scorecard and Mass Emission Quantification

Dual-Accounting Regulatory Approach

Under Decree 06-138, industrial emissions are strictly regulated by stack concentration limits (mg/Nm³), while corporate sustainability frameworks require total mass accounting (metric tonnes). Both metrics are quantified and cross-verified below.

Pollutant	Code	HBNS (t)	El Merk (t)	Total (t)	HBNS Conc.	El Merk Conc.	Limit	Status
Pollutant	Formula	HBNS Mass (t)	El Merk Mass (t)	Total Mass (t)	HBNS Stack Conc.	El Merk Stack Conc.	Decree 06-138 Limit	Compliance Verdict
Nitrogen Dioxide	NO2	1,420.0	1,890.0	3,310.0	788.0 mg/Nm³	980.8 mg/Nm³	200 mg/Nm³	EXCEEDED (Retest)
Carbon Monoxide	CO	310.0	420.0	730.0	105.0 mg/Nm³	118.4 mg/Nm³	150 mg/Nm³	COMPLIANT
Sulfur Dioxide	SO2	18.5	24.2	42.7	12.0 mg/Nm³	15.3 mg/Nm³	800 mg/Nm³	COMPLIANT
Particulate Matter	PM	22.0	29.5	51.5	8.5 mg/Nm³	11.2 mg/Nm³	30 mg/Nm³	COMPLIANT
Volatile Organic Cpds	VOC	115.0	145.0	260.0	42.0 mg/Nm³	51.6 mg/Nm³	150 mg/Nm³	COMPLIANT

Analysis of Decree 06-138 Compliance Findings:

- NO2 Concentration Thresholds: Both HBNS (788 mg/Nm³) and El Merk (980.8 mg/Nm³) exceed the 200 mg/Nm³ limit for legacy industrial gas turbines under high mechanical load. A joint Sonatrach engineering team has initiated Dry Low NOx (DLN) burner retrofit studies and scheduled mandatory stack re-testing per Ministry guidelines.
- CO, SO2, PM, VOC Full Compliance: Carbon monoxide levels (105-118 mg/Nm³) are well below the 150 mg/Nm³ ceiling, reflecting excellent fuel gas combustion efficiency. Sulfur dioxide concentrations (12-15 mg/Nm³) remain trace due to the ultra-sweet nature of Berkine formation gas.

CHAPTER 8
 MULTI-METRIC INTENSITIES & OGCI BENCHMARKING

Dual-Denominated Carbon Intensities, Methane Metrics, and International Comparisons

Metric	Unit	HBNS	El Merk	Total	OGCI Target	Status
Intensity Metric	Unit	HBNS	El Merk	Consolidated Total	OGCI Upstream Target	Performance Status
Carbon Intensity (Total BOE)	kg CO2e / Total BOE	11.18	18.49	14.58	17.00 kg/BOE	WITHIN TARGET
Carbon Intensity (Saleable BOE)	kg CO2e / Saleable BOE	30.84	36.05	33.72	— (Internal Benchmark)	-6.2% YoY Improvement
Methane Intensity (NGSI wt.%)	wt.% of Gas Mass	0.034%	0.055%	0.043%	< 0.200 wt.%	TOP DECILE PERFORMANCE
Flaring Intensity (Volume/BOE)	Sm3 / Total BOE	0.68	1.15	0.90	—	-28.1% vs Baseline

The Dual-Denominator Accounting Rationale

Total BOE reflects overall operational engineering efficiency including heavy gas re-injection compressors (achieving 14.58 kg CO2e/BOE, well below OGCI 17.0 ceiling). Saleable BOE reflects carbon embedded per exported commercial product (33.72 kg CO2e/BOE), providing transparent dual-perspective accounting.

5-Year Carbon Intensity Trajectory (2021–2025):

Metric (kg CO2e / BOE)	2021	2022	2023	2024	2025	5-Year Change
Carbon Intensity (Total BOE)	17.44	17.48	16.84	14.79	14.58	-16.4%
Carbon Intensity (Saleable BOE)	35.94	36.84	35.88	33.99	33.72	-6.2%
Methane Intensity (NGSI wt.%)	0.165%	0.160%	0.147%	0.072%	0.043%	-73.9%

CHAPTER 9

JOINT VENTURE PARTNER EQUITY SHARE ALLOCATIONS

Apportionment of Verified Scope 1 GHG and Methane Emissions per International Ownership

Equity Share Accounting Standard (GHG Protocol)

Under the equity share approach, Groupement Berkine emissions are allocated according to economic interest in the joint venture. As designated national operator, Sonatrach holds a majority 51.0% stake across both concessions.

JV Partner Name	Ticker / Code	Country	Equity Share (%)	Operator Status	Allocated Scope 1 (tCO2e)	Allocated CH4 (Tonnes)
Sonatrach (National Company)	SH	Algeria	51.00%	Designated Operator	973,437.00	1,713.60
Occidental Petroleum Corp.	OXY	United States	24.50%	Non-Operating Partner	467,631.50	823.20
Eni S.p.A.	ENI	Italy	12.25%	Non-Operating Partner	233,815.75	411.60
TotalEnergies SE	TTE	France	4.90%	Non-Operating Partner	93,526.30	164.64
Pertamina	PER	Indonesia	4.90%	Non-Operating Partner	93,526.30	164.64
Repsol S.A.	REP	Spain	2.45%	Non-Operating Partner	46,763.15	82.32
Total Consolidated JV Allocation	100.00%	Global	100.00%	Consortium	1,908,700.00	3,360.00

Application of Equity Share Disclosures for Partner CSRD & SEC Reporting:

- Occidental Petroleum: 467.6 ktCO2e allocated for inclusion in US SEC Scope 1 equity reporting.
- Eni & TotalEnergies: 233.8 ktCO2e and 93.5 ktCO2e respectively integrated into European CSRD/ESRS E1 verified non-financial reporting.
- Time-Sliced Consistency: Historical equity adjustments resulting from concession renewals are dynamically accounted for per contract effective dates.

CHAPTER 10 & 11
 DECARBONIZATION PROJECTS & QAI/QC VERIFICATION

Decarbonization Capital Projects and ISO 14064-1 Internal Verification Procedures

Project Title	Facility	Target Emission Source	Est. Reduction (tCO2e/yr)	Completion	Capex Status
Flare Gas Recovery System (FGRS)	EI Merk	Routine & Safety Flaring	65,000	Q3 2026	Under Construction
High-Pressure Gas Re-injection Upgrade	HBNS	Venting & Relief Systems	25,000	Completed	Operational
Optical Gas Imaging (OGI) LDAR Program	Both	Fugitive Equipment Leaks	8,200	Active Continuous	Operational
Gas Turbine Dry Low NOx (DLN) Upgrade	EI Merk	NOx Combustion Emissions	N/A (CAP focus)	Q4 2026	Engineering Study
Low-Pressure Flare Sweep Gas Minimization	HBNS	Purge Gas Combustion	12,500	Completed	Operational

Chapter 11: Quality Assurance & Quality Control (QA/QC) Verification

Verification Category	Control Procedure & ISO 14064-1 Specification	Audit Frequency	2025 Audit Finding
Custody Transfer Metering	Calibrate Coriolis and ultrasonic flare meters against API MPMS 14	Semi-Annual	Zero drift detected ($\pm 0.4\%$ error)
Fuel Gas Chromatography	Weekly gas sampling for C1-C6+ speciation, molecular weight, HHV	Weekly	Stoichiometry aligned with API Compendium
VISR Optical Surveillance	Continuous multispectral infrared tracking of flare flame envelope	Continuous 24/7	Combustion efficiency verified >99.85%
LDAR Fugitive Inspections	Survey 100% of flanges, valves, pumps using FLIR GFx320 cameras	Quarterly	38 minor leaks identified and repaired

CHAPTER ANNEX A

5-YEAR CONSOLIDATED HISTORICAL PERFORMANCE (2021–2025)

Multi-Year Production Profiles, Flaring Evolutions, and Performance Intensities

Table A.1: 5-Year Hydrocarbon Production Profile

Product Stream	Unit	2021	2022	2023	2024	2025
Gross Gas Production	MMSm3	8,743.69	9,462.00	9,786.27	10,077.56	9,757.51
Injected Gas Volume	MMSm3	7,639.83	8,272.99	8,644.16	9,164.25	8,990.01
Gas w/o Injected Gas	MMSm3	1,103.86	1,189.01	1,142.11	913.31	767.50
Liquid Production (Crude/LPG)	MMBOE	57.77	59.16	59.97	56.14	55.38
Total Production (Gross)	MMBOE	126.30	132.89	135.89	134.28	130.93
Total Production w/o Injected Gas	MMBOE	66.56	68.59	69.00	63.26	61.27
Total Saleable Production	MMBOE	61.28	63.06	63.76	58.47	56.62

Table A.2: 5-Year Operational Flaring Streams & Regulatory Compliance

Flaring Category	Unit	2021	2022	2023	2024	2025
Routine Flaring	kNm3	100,396	83,366	81,771	77,291	73,986
Non-Routine Flaring	kNm3	3,370	2,682	3,425	30,007	36,369
Safety & Purge Flaring	kNm3	87,596	106,967	53,952	6,706	7,543
Total Flared Volume	kNm3	191,362	193,015	139,148	114,004	117,898
Flaring Intensity (% Gross Gas)	vol.%	2.19%	2.04%	1.42%	1.13%	1.21%
HBNS Asset Intensity	vol.%	1.92%	1.81%	1.24%	0.98%	0.86%
Decree 21-330 Status (HBNS)	Limit "d 1.00%	Exceeded	Exceeded	Exceeded	COMPLIANT	COMPLIANT

CHAPTER ANNEX A (CONT.)

5-YEAR PERFORMANCE INTENSITIES & CAP MASS EVOLUTION

Multi-Year Historical Trends in Carbon, Methane, and Criteria Pollutants

Table A.3: 5-Year Carbon & Methane Intensities

Performance Metric	Unit	2021	2022	2023	2024	2025
Carbon Intensity (Total BOE)	kg CO2e / BOE	17.44	17.48	16.84	14.79	14.58
Carbon Intensity (Saleable BOE)	kg CO2e / BOE	35.94	36.84	35.88	33.99	33.72
Methane Intensity (NGSI)	wt. %	0.165%	0.160%	0.147%	0.072%	0.043%
Flaring Volume per BOE	Sm3 / BOE	3.24	3.37	2.14	2.33	2.15

Table A.4: 5-Year Criteria Air Pollutants Mass Evolution (Tonnes/Year)

Criteria Pollutant	Unit	2021	2022	2023	2024	2025
Nitrogen Dioxide (NO2)	Tonnes	3,840.0	3,910.0	3,780.0	3,450.0	3,310.0
Carbon Monoxide (CO)	Tonnes	860.0	890.0	840.0	760.0	730.0
Sulfur Dioxide (SO2)	Tonnes	52.0	54.0	49.0	45.0	42.7
Particulate Matter (PM)	Tonnes	64.0	66.0	61.0	54.0	51.5
Volatile Organic Cpds (VOC)	Tonnes	340.0	360.0	330.0	280.0	260.0
Total Criteria Pollutants Mass	Tonnes	5,156.0	5,280.0	5,060.0	4,589.0	4,394.2

CHAPTER ANNEX B

VERIFIED EMISSIONS ACTIVITY LEDGER & FORMAL SIGN-OFF

Sample Operational Logs and Executive Verification Certificate

Month	Facility	Scope	Source Module	Fuel Type	Activity Qty	Unit	CO2 (t)	CH4 (t)	tCO2e
2025-12	HBNS (Block 404a)	Scope 1	Stationary Combustion	Field Natural Gas	42,100	kNm3	55,175.0	123.3	58,627.4
2025-12	HBNS (Block 404a)	Scope 1	Routine Flaring	Sweet Flare Gas	2,228	kNm3	9,875.0	10.2	10,160.6
2025-12	HBNS (Block 404a)	Scope 1	Non-Routine Flaring	Depressurization Gas	1,591	kNm3	7,050.0	7.3	7,254.4
2025-12	HBNS (Block 404a)	Scope 1	Fugitive Equipment Leaks	Fugitive Gas	—	—	0.0	4.5	126.0
2025-12	El Merk (Block 208)	Scope 1	Stationary Combustion	High-Pressure Gas	60,400	kNm3	78,941.6	156.7	83,329.2
2025-12	El Merk (Block 208)	Scope 1	Routine Flaring	Separator LP Gas	3,937	kNm3	14,325.0	15.1	14,747.8
2025-12	El Merk (Block 208)	Scope 1	Non-Routine Flaring	Compressor Trip Gas	1,439	kNm3	5,230.0	5.5	5,384.0
2025-12	El Merk (Block 208)	Scope 1	Fugitive Equipment Leaks	Fugitive Gas	—	—	0.0	4.5	126.0

EXECUTIVE VERIFICATION SIGN-OFF & CERTIFICATION

This quantified inventory has been prepared in accordance with the mandatory provisions of ISO 14064-1:2018, the GHG Protocol Corporate Standard, and Algerian Executive Decrees 21-330 and 06-138. The reported activity data, continuous optical gas imaging DRE measurements, and emissions estimates represent a true and fair view of Groupement Berkine operational performance for the fiscal year 2025.

Lead MRV Engineer: HSE & Decarbonization Division
 Managing Director: Groupement Berkine Executive Committee